MIDTERM TASK 1 - Data Cleaning and Preparation

This task focuses on preparing raw data for analysis by identifying and correcting errors, inconsistencies, and missing values. Students will clean the dataset by removing duplicates, handling incomplete or incorrect entries, and formatting the data into a consistent structure. The goal is to produce a well-organized and reliable dataset that is ready for further analysis and visualization.

Bike Sales Fact table

Sales_Order #	Date	Day	Month	Year	Customer_Age	Age_Group	Customer_Gender	Country	State	Product_Category	Sub_Category	Model	Color	Size	Order_Quantity	Unit_Cost	Unit_Price	Profit	Cost
000261695	01/12/2021	1	December	2021	39	Adults (35-64)	Female	United States	California	Bikes	Mountain Bikes	Mountain-200	Black	46	4	$1,252.00	$2,295.00	$4,172.00	$5,008.00
000261696	01/12/2021	1	December	2021	44	Adults (35-64)	Male	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200	Silver	42	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00
000261697	02/12/2021	2	December	2021	37	Adults (35-64)	Male	United States	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	2	$420.00	$769.00	$698.00	$840.00
000261698	02/12/2021	2	December	2021	31	Young Adults (25-34)	Female	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	42	1	$420.00	$769.00	$349.00	$420.00
000261699	03/12/2021	3	December	2021	37	Adults (35-64)	Female	United States	California	Bikes	Mountain Bikes	Mountain-200	Black	46	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00
000261700	03/12/2021	3	December	2021	24	Youth (<25)	Female	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200	Black	38	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00
000261701	03/12/2021	3	December	2021	37	Adults (35-64)	Male	United States	Washington	Bikes	Mountain Bikes	Mountain-200	Black	46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00
000261702	04/12/2021	4	December	2021	31	Young Adults (25-34)	Female	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	42	4	$420.00	$769.00	$1,396.00	$1,680.00
000261703	05/12/2021	5	December	2021	39	Adults (35-64)	Female	United States	California	Bikes	Mountain Bikes	Mountain-200	Black	46	4	$1,252.00	$2,295.00	$4,172.00	$5,008.00
000261704	05/12/2021	5	December	2021	42	Adults (35-64)	Male	Germany	Nordrhein-Westfalen	Bikes	Mountain Bikes	Mountain-200	Black	38	4	$1,252.00	$2,295.00	$4,172.00	$5,008.00
000261705	05/12/2021	5	December	2021	35	Adults (35-64)	Female	Australia	Queensland	Bikes	Mountain Bikes	Mountain-200	Silver	38	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00
000261706	05/12/2021	5	December	2021	37	Adults (35-64)	Female	United States	California	Bikes	Mountain Bikes	Mountain-200	Black	46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00
000261707	06/12/2021	6	December	2021	23	Youth (<25)	Male	United Kingdom	England	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	3	$420.00	$769.00	$1,047.00	$1,260.00
000261708	06/12/2021	6	December	2021	27	Young Adults (25-34)	Male	Canada	British Columbia	Bikes	Mountain Bikes	Mountain-200	Black	46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00
000261709	06/12/2021	6	December	2021	36	Youth (<25)	Male	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	42	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00
000261710	06/12/2021	6	December	2021	47	Adults (35-64)	Male	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200	Silver	38	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00
000261711	07/12/2021	7	December	2021	30	Young Adults (25-34)	Male	United States	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	38	4	$420.00	$769.00	$1,396.00	$1,680.00
000261712	07/12/2021	7	December	2021	38	Adults (35-64)	Male	United States	California	Bikes	Mountain Bikes	Mountain-200	Silver	42	2	$1,266.00	$2,320.00	$2,108.00	$2,532.00
000261713	08/12/2021	8	December	2021	19	Youth (<25)	Female	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-500	Silver	42	4	$308.00	$565.00	$1,028.00	$1,232.00
000261714	08/12/2021	8	December	2021	30	Young Adults (25-34)	Female	Canada	British Columbia	Bikes	Mountain Bikes	Mountain-200	Silver	38	4	$1,266.00	$2,320.00	$4,216.00	$5,064.00
000261715	08/12/2021	8	December	2021	39	Adults (35-64)	Female	United States	Oregon	Bikes	Mountain Bikes	Mountain-200	Black	42	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00
000261716	08/12/2021	8	December	2021	35	Adults (35-64)	Female	United States	California	Bikes	Mountain Bikes	Mountain-500	Black	42	4	$295.00	$540.00	$245.00	$1,180.00
000261717	09/12/2021	9	December	2021	33	Young Adults (25-34)	Female	Australia	Victoria	Bikes	Mountain Bikes	Mountain-100	Black	38	2	$1,898.00	$3,375.00	$2,954.00	$3,796.00
000261718	09/12/2021	9	December	2021	41	Adults (35-64)	Female	Germany	Hamburg	Bikes	Mountain Bikes	Mountain-200	Silver	42	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00
000261719	10/12/2021	10	December	2021	34	Young Adults (25-34)	Female	United States	California	Bikes	Mountain Bikes	Mountain-200	Black	42	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00
000261720	10/12/2021	10	December	2021	40	Adults (35-64)	Male	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	42	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00
000261721	10/12/2021	10	December	2021	26	Young Adults (25-34)	Male	United Kingdom	England	Bikes	Mountain Bikes	Mountain-200	Black	38	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00
000261722	10/12/2021	10	December	2021	34	Young Adults (25-34)	Male	United States	California	Bikes	Mountain Bikes	Mountain-500	Black	40	1	$295.00	$540.00	$245.00	$295.00
000261723	10/12/2021	10	December	2021	34	Young Adults (25-34)	Female	United States	Washington	Bikes	Mountain Bikes	Mountain-100	Silver	44	1	$1,912.00	$3,400.00	$1,488.00	$1,912.00
000261724	10/12/2021	10	December	2021	38	Adults (35-64)	Male	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	38	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00
000261725	11/12/2021	11	December	2021	24	Youth (<25)	Female	France	Seine (Paris)	Bikes	Mountain Bikes	Mountain-200	Black	38	3	$1,252.00	$2,295.00	$3,129.00	$3,756.00
000261726	11/12/2021	11	December	2021	41	Adults (35-64)	Female	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-400-W	Silver	38	2	$420.00	$769.00	$698.00	$840.00
000261727	11/12/2021	11	December	2021	27	Young Adults (25-34)	Male	Canada	British Columbia	Bikes	Mountain Bikes	Mountain-200	Black	46	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00
000261728	11/12/2021	11	December	2021	37	Adults (35-64)	Male	United States	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	1	$420.00	$769.00	$349.00	$420.00
000261729	11/12/2021	11	December	2021	38	Adults (35-64)	Female	United States	California	Bikes	Mountain Bikes	Mountain-200	Silver	38	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00
000261730	12/12/2021	12	December	2021	36	Adults (35-64)	Female	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200	Silver	42	4	$1,266.00	$2,320.00	$4,216.00	$5,064.00
000261731	12/12/2021	12	December	2021	37	Adults (35-64)	Male	United States	California	Bikes	Mountain Bikes	Mountain-400-W	Silver	46	4	$420.00	$769.00	$1,396.00	$1,680.00
000261732	12/12/2021	12	December	2021	34	Young Adults (25-34)	Male	Australia	New South Wales	Bikes	Mountain Bikes	Mountain-200	Black	38	2	$1,252.00	$2,295.00	$2,086.00	$2,504.00

BIKE_SALES ORIG | SALES_ORDER | CUSTOMERS | COUNTRY | PRODUCTS | PRODUCT_SUB_CATEGORY

Sales order dim table

Sales_Order #	Day	Month	Year	Customer_id	Product_id	Order_Quantity	Unit_Cost	Unit_Price	Profit	Cost	Revenue
000261695	1	December	2021	c-101	p-105	4	$1,252.00	$2,295.00	$2,954.00	$5,008.00	$9,180.00
000261696	1	December	2021	c-102	p-116	1	$1,266.00	$2,320.00	$5,908.00	$1,266.00	$2,320.00
000261697	2	December	2021	c-103	p-126	2	$420.00	$769.00	$5,908.00	$840.00	$1,538.00
000261698	2	December	2021	c-104	p-127	1	$420.00	$769.00	$1,488.00	$420.00	$769.00
000261699	3	December	2021	c-105	p-106	2	$1,252.00	$2,295.00	$4,172.00	$2,504.00	$4,590.00
000261700	3	December	2021	c-106	p-107	1	$1,252.00	$2,295.00	$2,086.00	$1,252.00	$2,295.00
000261701	3	December	2021	c-107	p-108	1	$1,252.00	$2,295.00	$1,043.00	$1,252.00	$2,295.00
000261702	4	December	2021	c-104	p-128	4	$420.00	$769.00	$1,043.00	$1,680.00	$3,076.00
000261703	5	December	2021	c-101	p-105	4	$1,252.00	$2,295.00	$4,172.00	$5,008.00	$9,180.00
000261704	5	December	2021	c-108	p-109	4	$1,252.00	$2,295.00	$1,043.00	$5,008.00	$9,180.00
000261705	5	December	2021	c-109	p-117	1	$1,266.00	$2,320.00	$2,086.00	$1,266.00	$2,320.00
000261706	5	December	2021	c-105	p-108	1	$1,252.00	$2,295.00	$3,129.00	$1,252.00	$2,295.00
000261707	6	December	2021	c-110	p-129	3	$420.00	$769.00	$2,086.00	$1,260.00	$2,307.00
000261708	6	December	2021	c-111	p-108	1	$1,252.00	$2,295.00	$3,129.00	$1,252.00	$2,295.00
000261709	6	December	2021	c-112	p-110	1	$1,252.00	$2,295.00	$4,172.00	$1,252.00	$2,295.00
000261710	6	December	2021	c-113	p-117	1	$1,266.00	$2,320.00	$1,054.00	$1,266.00	$2,320.00
000261711	7	December	2021	c-114	p-130	4	$420.00	$769.00	$1,054.00	$1,680.00	$3,076.00
000261712	7	December	2021	c-115	p-118	2	$1,266.00	$2,320.00	$2,108.00	$2,532.00	$4,640.00
000261713	8	December	2021	c-116	p-140	4	$308.00	$565.00	$4,216.00	$1,232.00	$2,260.00
000261714	8	December	2021	c-117	p-119	4	$1,266.00	$2,320.00	$4,216.00	$5,064.00	$9,280.00
000261715	8	December	2021	c-118	p-111	2	$1,252.00	$2,295.00	$3,162.00	$2,504.00	$4,590.00
000261716	8	December	2021	c-119	p-135	4	$295.00	$540.00	$2,108.00	$1,180.00	$2,160.00
000261717	9	December	2021	c-120	p-101	2	$1,898.00	$3,375.00	$1,054.00	$3,796.00	$6,750.00
000261718	9	December	2021	c-121	p-116	1	$1,266.00	$2,320.00	$4,216.00	$1,266.00	$2,320.00
000261719	10	December	2021	c-122	p-111	2	$1,252.00	$2,295.00	$3,162.00	$2,504.00	$4,590.00
000261720	10	December	2021	c-123	p-111	2	$1,252.00	$2,295.00	$698.00	$2,504.00	$4,590.00
000261721	10	December	2021	c-124	p-107	1	$1,252.00	$2,295.00	$349.00	$1,252.00	$2,295.00
000261722	10	December	2021	c-125	p-136	1	$295.00	$540.00	$1,396.00	$295.00	$540.00
000261723	10	December	2021	c-126	p-104	1	$1,912.00	$3,400.00	$1,047.00	$1,912.00	$3,400.00
000261724	10	December	2021	c-127	p-107	1	$1,252.00	$2,295.00	$1,396.00	$1,252.00	$2,295.00
000261725	11	December	2021	c-128	p-112	3	$1,252.00	$2,295.00	$698.00	$3,756.00	$6,885.00
000261726	11	December	2021	c-129	p-131	2	$420.00	$769.00	$349.00	$840.00	$1,538.00
000261727	11	December	2021	c-111	p-108	1	$1,252.00	$2,295.00	$1,396.00	$1,252.00	$2,295.00
000261728	11	December	2021	c-103	p-132	1	$420.00	$769.00	$1,047.00	$420.00	$769.00
000261729	11	December	2021	c-130	p-117	1	$1,266.00	$2,320.00	$245.00	$1,266.00	$2,320.00
000261730	12	December	2021	c-131	p-120	4	$1,266.00	$2,320.00	$245.00	$5,064.00	$9,280.00
000261731	12	December	2021	c-103	p-133	4	$420.00	$769.00	$980.00	$1,680.00	$3,076.00
000261732	12	December	2021	c-132	p-113	2	$1,252.00	$2,295.00	$490.00	$2,504.00	$4,590.00

BIKE_SALES ORIG | SALES_ORDER | CUSTOMERS | COUNTRY | PRODUCTS | PRODUCT_SUB_CATEGORY

Customer dim table

Customer_id	Customer_Age	Age_Group	Customer_Gender	State	Country_id
c-101	39	Adults (35-64)	Female	California	co-101
c-102	44	Adults (35-64)	Male	England	co-102
c-103	37	Adults (35-64)	Male	California	co-101
c-104	31	Young Adults (25-34)	Female	New South Wales	co-103
c-105	37	Adults (35-64)	Female	California	co-101
c-106	24	Youth (<25)	Female	England	co-102
c-107	37	Adults (35-64)	Male	Washington	co-101
c-108	42	Adults (35-64)	Male	Nordrhein-Westfalen	co-104
c-109	35	Adults (35-64)	Female	Queensland	co-103
c-110	23	Youth (<25)	Male	England	co-102
c-111	27	Young Adults (25-34)	Male	British Columbia	co-105
c-112	36	Youth (<25)	Male	New South Wales	co-103
c-113	47	Adults (35-64)	Male	England	co-102
c-114	30	Young Adults (25-34)	Male	California	co-101
c-115	38	Adults (35-64)	Male	California	co-101
c-116	19	Youth (<25)	Female	New South Wales	co-103
c-117	30	Young Adults (25-34)	Female	British Columbia	co-105
c-118	39	Adults (35-64)	Female	Oregon	co-101
c-119	35	Adults (35-64)	Female	California	co-101
c-120	33	Young Adults (25-34)	Female	Victoria	co-103
c-121	41	Adults (35-64)	Female	Hamburg	co-104
c-122	34	Young Adults (25-34)	Female	California	co-101
c-123	40	Adults (35-64)	Male	New South Wales	co-103
c-124	26	Young Adults (25-34)	Male	England	co-102
c-125	34	Young Adults (25-34)	Male	California	co-101
c-126	34	Young Adults (25-34)	Female	Washington	co-101
c-127	38	Adults (35-64)	Male	New South Wales	co-103
c-128	24	Youth (<25)	Female	Seine (Paris)	co-106
c-129	41	Adults (35-64)	Female	New South Wales	co-103
c-130	38	Adults (35-64)	Female	California	co-101
c-131	36	Adults (35-64)	Female	New South Wales	co-103
c-132	34	Young Adults (25-34)	Male	New South Wales	co-103
c-133	35	Adults (35-64)	Female	Victoria	co-103
c-134	38	Adults (35-64)	Female	Washington	co-101
c-135	32	Young Adults (25-34)	Female	Queensland	co-103
c-136	40	Adults (35-64)	Female	California	co-101
c-137	44	Adults (35-64)	Female	England	co-102

BIKE_SALES ORIG | SALES_ORDER | CUSTOMERS | COUNTRY | PRODUCTS | PRODUCT_SUB_CATEGORY

Country dim table

Country	Country_id
United States	co-101
United Kingdom	co-102
Australia	co-103
Germany	co-104
Canada	co-105
France	co-106

Product dim table

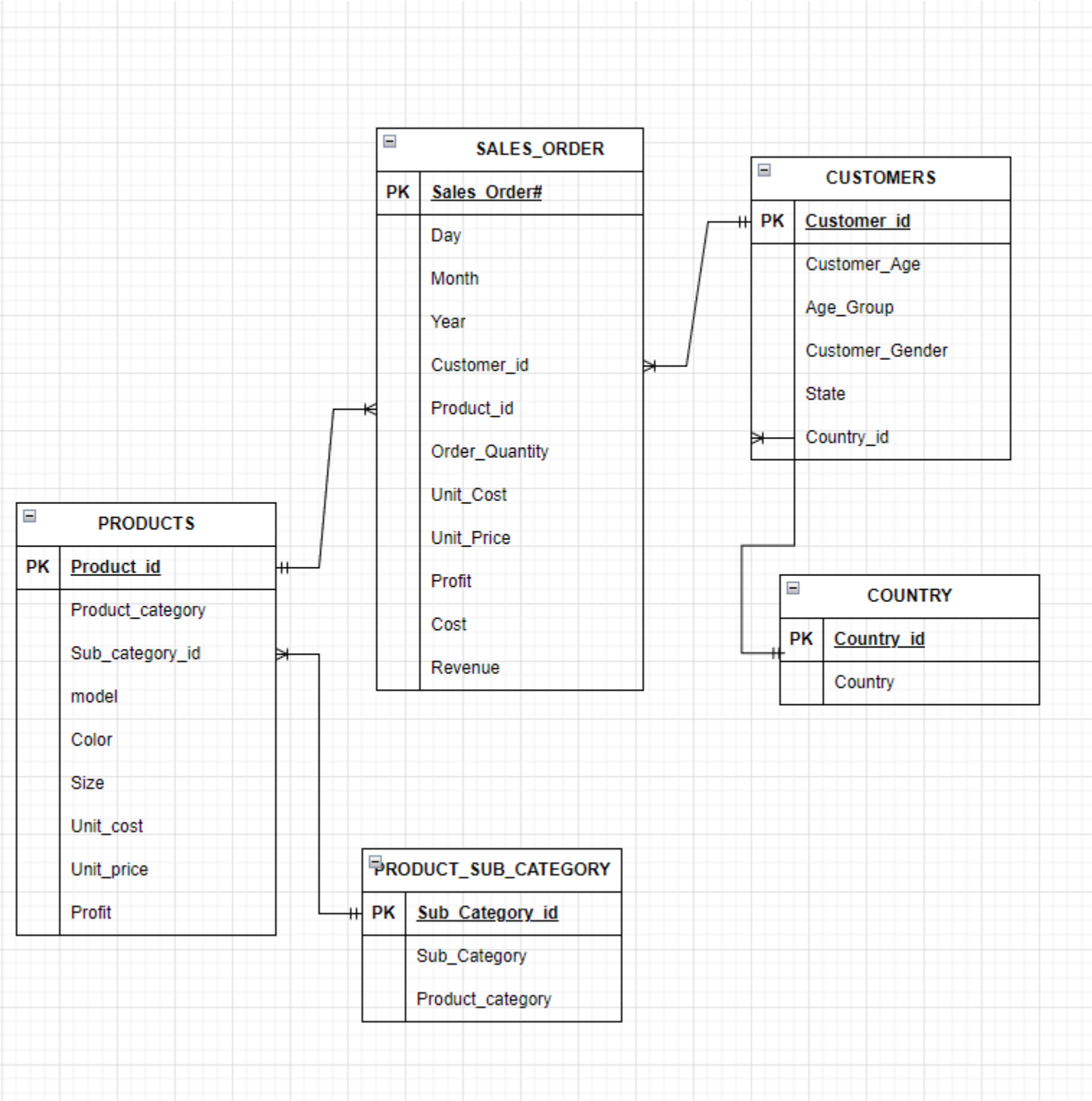
Product_id	Product_Category	Sub_Category_id	Model	Color	Size	Unit_Cost	Unit_Price	Profit
p-101	Bikes	sc-101	Mountain-100	Black	38	$1,898.00	$3,375.00	$2,954.00
p-102	Bikes	sc-101	Mountain-100	Black	48	$1,898.00	$3,375.00	$5,908.00
p-103	Bikes	sc-101	Mountain-100	Black	38	$1,898.00	$3,375.00	$5,908.00
p-104	Bikes	sc-101	Mountain-100	Silver	44	$1,912.00	$3,400.00	$1,488.00
p-105	Bikes	sc-101	Mountain-200	Black	46	$1,252.00	$2,295.00	$4,172.00
p-106	Bikes	sc-101	Mountain-200	Black	46	$1,252.00	$2,295.00	$2,086.00
p-107	Bikes	sc-101	Mountain-200	Black	38	$1,252.00	$2,295.00	$1,043.00
p-108	Bikes	sc-101	Mountain-200	Black	46	$1,252.00	$2,295.00	$1,043.00
p-109	Bikes	sc-101	Mountain-200	Black	38	$1,252.00	$2,295.00	$4,172.00
p-110	Bikes	sc-101	Mountain-200	Black	42	$1,252.00	$2,295.00	$1,043.00
p-111	Bikes	sc-101	Mountain-200	Black	42	$1,252.00	$2,295.00	$2,086.00
p-112	Bikes	sc-101	Mountain-200	Black	38	$1,252.00	$2,295.00	$3,129.00
p-113	Bikes	sc-101	Mountain-200	Black	38	$1,252.00	$2,295.00	$2,086.00
p-114	Bikes	sc-101	Mountain-200	Black	42	$1,252.00	$2,295.00	$3,129.00
p-115	Bikes	sc-101	Mountain-200	Black	42	$1,252.00	$2,295.00	$4,172.00
p-116	Bikes	sc-101	Mountain-200	Silver	42	$1,266.00	$2,320.00	$1,054.00
p-117	Bikes	sc-101	Mountain-200	Silver	38	$1,266.00	$2,320.00	$1,054.00
p-118	Bikes	sc-101	Mountain-200	Silver	42	$1,266.00	$2,320.00	$2,108.00
p-119	Bikes	sc-101	Mountain-200	Silver	38	$1,266.00	$2,320.00	$4,216.00
p-120	Bikes	sc-101	Mountain-200	Silver	42	$1,266.00	$2,320.00	$4,216.00
p-121	Bikes	sc-101	Mountain-200	Silver	42	$1,266.00	$2,320.00	$3,162.00
p-122	Bikes	sc-101	Mountain-200	Silver	38	$1,266.00	$2,320.00	$2,108.00
p-123	Bikes	sc-101	Mountain-200	Silver	46	$1,266.00	$2,320.00	$1,054.00
p-124	Bikes	sc-101	Mountain-200	Silver	46	$1,266.00	$2,320.00	$4,216.00
p-125	Bikes	sc-101	Mountain-200	Silver	38	$1,266.00	$2,320.00	$3,162.00
p-126	Bikes	sc-101	Mountain-400-W	Silver	46	$420.00	$769.00	$698.00
p-127	Bikes	sc-101	Mountain-400-W	Silver	42	$420.00	$769.00	$349.00
p-128	Bikes	sc-101	Mountain-400-W	Silver	42	$420.00	$769.00	$1,396.00
p-129	Bikes	sc-101	Mountain-400-W	Silver	46	$420.00	$769.00	$1,047.00
p-130	Bikes	sc-101	Mountain-400-W	Silver	38	$420.00	$769.00	$1,396.00
p-131	Bikes	sc-101	Mountain-400-W	Silver	38	$420.00	$769.00	$698.00
p-132	Bikes	sc-101	Mountain-400-W	Silver	46	$420.00	$769.00	$349.00
p-133	Bikes	sc-101	Mountain-400-W	Silver	46	$420.00	$769.00	$1,396.00
p-134	Bikes	sc-101	Mountain-400-W	Silver	42	$420.00	$769.00	$1,047.00
p-135	Bikes	sc-101	Mountain-500	Black	42	$295.00	$540.00	$245.00
p-136	Bikes	sc-101	Mountain-500	Black	40	$295.00	$540.00	$245.00
p-137	Bikes	sc-101	Mountain-500	Black	44	$295.00	$540.00	$980.00

BIKE_SALES ORIG | SALES_ORDER | CUSTOMERS | COUNTRY | PRODUCTS | PRODUCT_SUB_CATEGORY

product_sub _category dim table

Sub_Category	Sub_Category_id	Product_Category
Mountain Bikes	sc-101	Bikes

draw.io table relationships diagram


SALES_ORDER
PK
Sales_Order#
Day
Month
Year
Customer_id
Product_id
Order_Quantity
Unit_Cost
Unit_Price
Profit
Cost
Revenue
CUSTOMERS
PK
Customer_id
Customer_Age
Age_Group
Customer_Gender
State
Country_id
PRODUCTS
PK
Product_id
Product_category
Sub_category_id
model
Color
Size
Unit_cost
Unit_price
Profit
COUNTRY
PK
Country_id
Country
PRODUCT_SUB_CATEGORY
PK
Sub_Category_id
Sub_Category
Product_category